AV AI VELOCITY ARCHITECT INTELLIGENCE · ISSUE 07

WEEK OF 10–16 OCTOBER 2026

The Sibling That Failed

Two tool calls ran in one batch. One failed, the other wrote an order. The history forgot the order, and the resume wrote it again.

14	8	8	14
packages read	executed runs	papers read	models priced

Vijay Zharotia
Healthcare AI architecture

WHY AI VELOCITY

Agents that act inherit obligations.

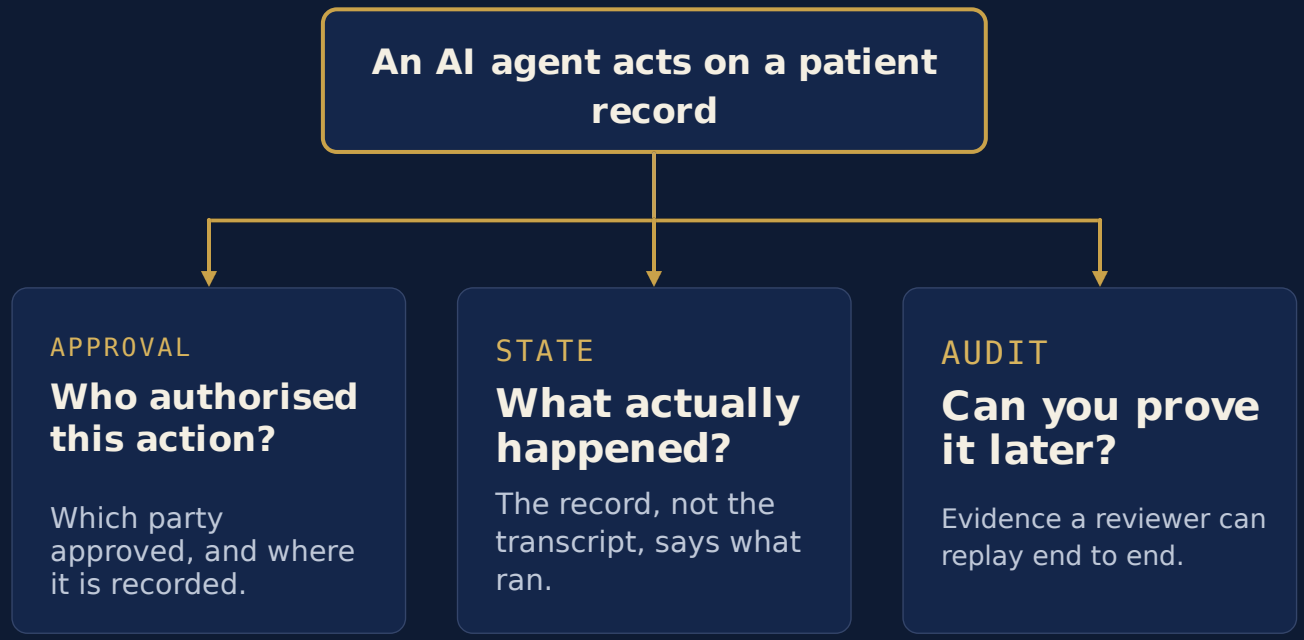
A weekly architecture brief for teams putting AI agents into healthcare workflows. We read the releases, run the mechanism, and say what to change.

FOR ARCHITECTS

The mechanism, the code, the move.

FOR PRODUCT MANAGERS

The decision, the risk, the date.



EVERY CLAIM CARRIES ITS EVIDENCE

EXECUTED

We ran it. The numbers are observed output.

DOCS

Read from documentation or source.

UNVERIFIED

Not confirmed. Vendor numbers carry an asterisk.

THE WEEK IN ONE VIEW

From a paused run to a failed batch

CONTEXT

Last week: a paused run

A paused run was state nobody read correctly. This week the unit became the batch: two tool calls in one model response, one of them a write.

→

PROBLEM

The batch failed as a unit

An apixaban order written beside a failed allergy lookup vanished from the history. The resume was told it never ran, and the record got a second one.

→

SOLUTION

Reconcile before you resume

Treat an ‘interrupted’ write as outcome unknown. Look it up in the record by idempotency key, then put the real result into the history.

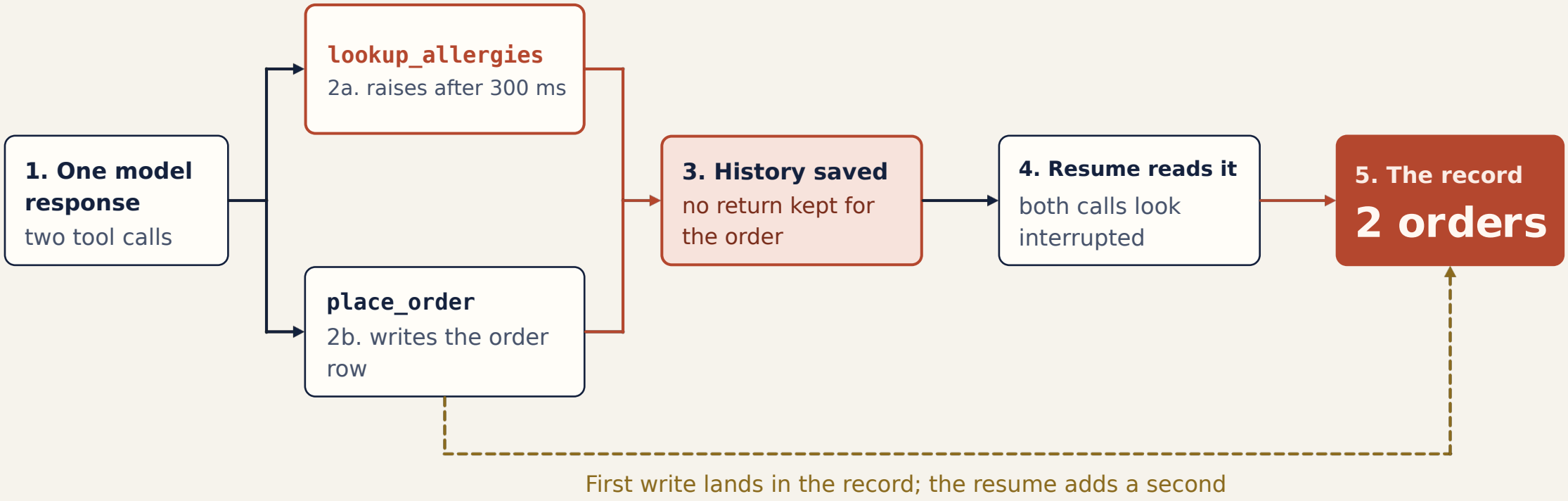
STATE · 04–06
What the history keeps

APPROVAL · 07
Who may write the approval

EVIDENCE · 08
Stand-ins for the record

COST, DATES · 09–10
Prices and deadlines

The order that ran twice



EXECUTED MECHANISM · SOLUTION · 2.55.0 AND YOUR HARNESS

Reconcile against the record before resuming

STEP 1
Run fails or is cancelled
Read ‘interrupted’ as outcome unknown.

→

STEP 2
Look up the write
By its idempotency key, in the record itself.

→

STEP 3
Repair the history
Put the real result in yourself.

→

STEP 4
Then resume
Only now may the run act again.

0 to 1
tool returns kept in the history after the failed run

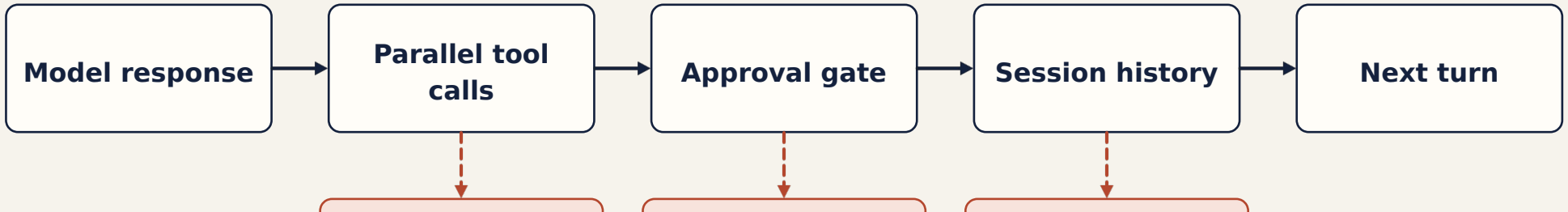
2 to 1
orders in the record after the resume

8 runs
and the model’s closing sentence was identical every time

> Pin pydantic-ai 2.55.0 or later explicitly: Python 3.10 resolves to 2.54.0. Histories saved before the upgrade are not repaired.

Three places a true fact left the history

CONTEXT
how it runs



PROBLEM
where it broke

OPENAI SDK 0.24
A sibling's failure dropped a finished output
DOCS

LANGCHAIN 1.4.6
Batched answers were matched by position
DOCS

LANGGRAPH 1.2.15
An invalid answer erased the open question
DOCS

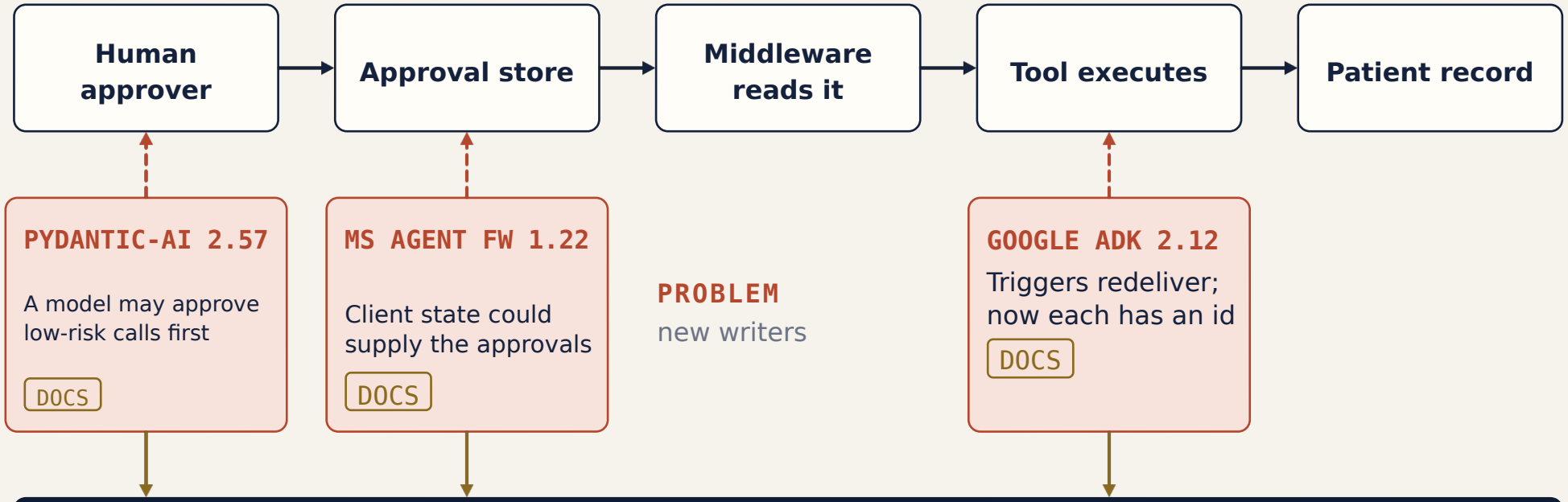
SOLUTION
what to build

Build each turn from a reconciled history

Check the history against the record before the next turn. Gate writes per call, re-raise invalid answers, and log what the reconciliation changed.

More parties can now write an approval

CONTEXT
how it runs



SOLUTION
what to build

Make the approver a recorded field

Store approver type, reviewer model id and version, and the delivery id with every write. Keep risk tiering in your code; keep record writes out of any model-approved tier.

Three stand-ins for evidence failed this week

CONTEXT
how it runs

Agent proposes a write → **A judge reviews it** → **History is compacted** → **Escalate or proceed** → **Write to the record**

PROBLEM
where it broke

2610.18497
7 of 7
judges favoured a visible wrong value

2610.20927
83%
of failed runs acted on an unsourced summary line

2610.21270
32–97%
of needed escalations missed on stated confidence

SOLUTION
what to build

Check claims against the record

Give any model reviewer an independently derived reference, keep only summary entries backed by a tool output, and make escalation an explicit workflow state.

COST · LIST PRICES READ 16 OCTOBER

Medical score 58 at $130 or at $650

MODEL	COST PER 10,000 DOCS	MEDICAL
Claude Sonnet 5.5	**$130**	**58**
Gemini 4 Argon	**$130; $260 from 1 Dec**	not shown
Claude Opus 5.5	**$260**	**61**
Claude Fable 5.1	**$650**	**58**
GPT-6 Astra	**$650**	**52**

Per million tokens in/out: Sonnet 5.5 and Argon $2/$10, Opus 5.5 $4/$20, Fable 5.1 and Astra $10/$50. 4,000 in and 500 out per document, no caching or retries. Argon rates per Google’s Gemini API docs.Medical = Artificial Analysis Healthcare & Medical. Argon is not shown there, so its Intelligence Index tie implies nothing here.

CONTEXT

A workload was validated on Argon at $130.

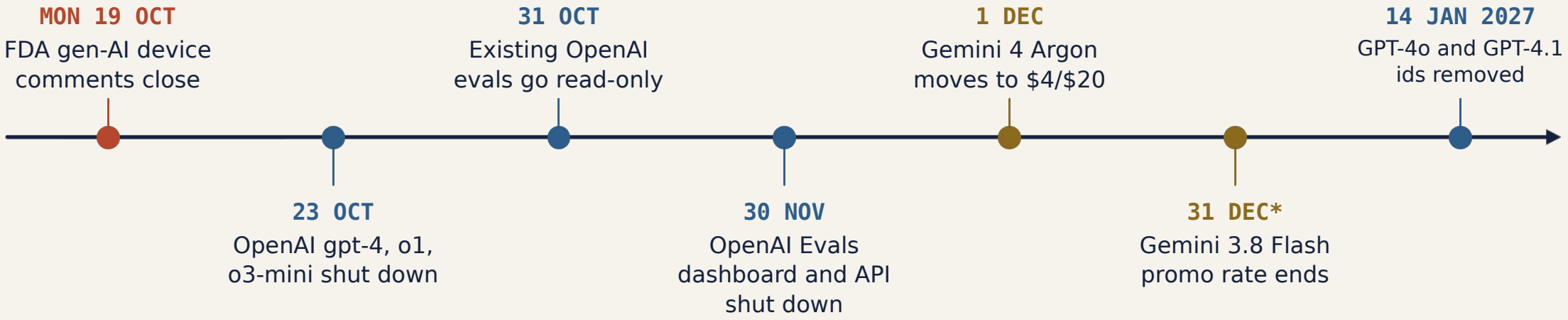
PROBLEM

From 1 December Google lists Argon at $4/$20, the same rate as Opus 5.5.

DECISION

For this workload shape, move to Sonnet 5.5 or pay double. Per task, reasoning tokens and cache hits can reorder the table: measure tokens per task on your own eval set first.

Dates for your change-control file



● REGULATORY ● PRICE ● MODEL RETIREMENT

*Third-party report; confirm on Google’s price page.

The approver is still the party without a field

ANTHROPIC

Tracing on by default

OTel spans when a tracer exists. Hooks now drop unknown keys, so a stamped approver id disappears.

OPENAI

Three new ways a session stops

Suspension, expiry and spending limits. None of them is a completion; handle each one.

ANACONDA

An agency score of 0 to 3

Level 3 needs per-action human approval, logged with a policy id and reason code.

GOOGLE

Argon GA, with a dated price

Gemini 4 Argon reached GA on 13 Oct; Google’s docs list $4/$20 from 1 Dec.

FDA

One deadline, nothing new

Comments on the generative-AI device paper close Monday 19 Oct.

QUIET THIS WEEK

MCP spec, Semantic Kernel, llama-index-core, HL7, CMS, ONC, OCR, NVIDIA healthcare. Open@Epic is 21–22 Oct.

Seven moves, with the date that forces each

Move	Why	By
Pin pydantic-ai 2.55.0+; reconcile resumes by idempotency key	Duplicate writes on resume	This sprint
Switch write-capable agents to per_call approvals	Answers matched by position	Next upgrade
Record approver type, model id and version on every approval	Models can now approve	Before enabling
Grep client state for the three approval list names	Agent Framework 1.22 drops them	Before upgrading
Choose Argon or Sonnet 5.5; write 1 Dec into validation	Argon intro rate ends	30 Nov
Write the GPT-4o and GPT-4.1 re-validation protocol	Five ids removed	14 Jan 2027
File an FDA comment if you hold a position	Docket FDA-2026-N-7874	Mon 19 Oct

AV AI VELOCITY

THIS WEEK'S POSITION

The record is the evidence. Everything else is a claim about it.

Long-form edition
Every mechanism, source and date

Lab package
Runs clean, no API key needed

New issue every Friday
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github.com/zkumar/ai-velocity-weekly